CSE 3230 • SOFTWARE DEVELOPMENT & WEB TECHNOLOGIES LAB

Testing & CI/CD Pipeline Guide

Complete Technical Walkthrough: 6 Unit Tests, 3 Integration Tests, and GitHub Actions CI

AUTHOR	APPLICATION	TEST FRAMEWORK	CI RUNNER
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1. Architectural Overview: How Testing Works in UMS

The University Management System (UMS) follows a classic enterprise three-tier Spring architecture secured by Spring Security. When demonstrating this system to your teacher, explain that automated tests exist at two distinct levels of the **Test Pyramid**:

```
Client / HTTP Request
        ↓
[ Spring Security Filter Chain ] — Authentication, Sessions, Roles (ADMIN, TEACHER, STUDENT)
        ↓
[ Controller Layer ] (TeacherController, StudentController, AdminController)
        ↓
[ Service Layer (Business Logic) ] (AdminService, StudentService, UserAccountService)
        ↓
[ JPA Repositories & Database ] (MySQL in production / H2 in-memory for testing)
```

UNIT TESTS — In Isolation

- **Target:** Service classes (`AdminService` , `StudentService`).
- **How:** Using **Mockito** to fake repositories.
- **Why:** Tests pure business rules without booting Spring or touching a database.
- **Speed:** Super fast (~**0.22 seconds** for 6 tests).

INTEGRATION TESTS — Full Slice

- **Target:** Controllers + Security + Database.
- **How:** Using **Spring MockMvc** + **in-memory H2 DB**.
- **Why:** Tests HTTP status codes, JSON validation, and security rules.
- **Speed:** Thorough (~**6.2 seconds** for the suite).

2. Deep Dive: The 6 Unit Tests (Mockito & JUnit 5)

Unit tests verify that business invariants are upheld even when invalid or malicious input is provided. Because these tests use `@ExtendWith(MockitoExtension.class)` , they run in milliseconds without starting Tomcat or Hibernate.

ID	Test File	Method Name	What It Tests & Why It Matters
UT-01	`AdminServiceTest`	`create_validAdmin_savesAndReturnsResponse`	Ensures that when an admin is created, password requirements and unique checks are executed before saving to the database.
UT-02	`AdminServiceTest`	`findById_nonExistingId_throwsResourceNotFoundException`	Ensures that requesting an unknown admin ID immediately throws a `ResourceNotFoundException` with the exact ID message.
UT-03	`AdminServiceTest`	`delete_lastAdmin_throwsConflictException`	**Critical Invariant:** If only 1 admin remains in the system, deleting it is blocked with a `ConflictException` to prevent permanent lockout.
UT-04	`StudentServiceTest`	`create_validStudent_savesAndReturnsResponse`	Validates full student creation flow: student ID uniqueness

ID	Test File	Method Name	What It Tests & Why It Matters
			check, department mapping, batch storage, and response DTO generation.
UT-05	`StudentServiceTest`	`create_duplicateStudentId_throwsConflictException`	Ensures two students cannot share the same institutional ID (e.g., `2007001`), throwing `ConflictException`.
UT-06	`StudentServiceTest`	`findById_nonExistingId_throwsResourceNotFoundException`	Validates that requesting a non-existent student ID triggers `ResourceNotFoundException` cleanly.

Code Walkthrough: Last Admin Protection Rule

```
// AdminServiceTest.java
@Test
void delete_lastAdmin_throwsConflictException() {
    Admin admin = new Admin();
    ReflectionTestUtils.setField(admin, "id", 1L); // Sets private ID field
    when(adminRepository.findById(1L)).thenReturn(Optional.of(admin));
    when(adminRepository.count()).thenReturn(1L); // Only 1 admin in system

    assertThatThrownBy(() -> adminService.delete(1L))
            .isInstanceOf(ConflictException.class)
            .hasMessage("Cannot delete the last admin account");

    // Verifies adminRepository.delete() was NEVER called
    verify(adminRepository, never()).delete(any());
}
```

> 💡 **Viva / Teacher Defense Tip #1: How did you set the entity ID without a setter?**
>
> **Teacher Question:** *"Your `AppUser` entity generates its ID automatically via JPA and has no `setId()` method. How did your unit test assign an ID to the mock entity?"*
>
> **Your Answer:** *"We used Spring's `ReflectionTestUtils.setField(admin, 'id', 1L)`. This safely populates the private field via reflection during unit testing, keeping our production domain model strictly encapsulated without exposing unnecessary setters."*

3. Deep Dive: The 3 Integration Tests (MockMvc & H2)

Integration tests execute real HTTP requests through the Spring Security filter chain, JSON controllers, validation framework, and an in-memory H2 database running in MySQL compatibility mode (`jdbc:h2:mem:testdb;MODE=MySQL`).

ID	Test File	Method Name	Scenario & Verification
IT-01	`TeacherIntegrationTest`	`fullTeacherLifecycle_createReadUpdateDelete`	Performs the complete REST lifecycle with credentials: • `POST /api/teachers` → **201 Created** + Location header • `GET /api/teachers/{id}` → **200 OK** • `PUT /api/teachers/{id}` → **200 OK** (updates designation) • `DELETE /api/teachers/{id}` → **204 No Content** • `GET /api/teachers/{id}` → **404 Not Found**
IT-02	`TeacherIntegrationTest`	`createTeacher_duplicateUsername_returnsConflict`	Submits two consecutive creations with username `hasan`. Confirms that the second returns **409 Conflict** with JSON error message: `Username 'hasan' is already taken`.
IT-03	`TeacherIntegrationTest`	`unauthenticatedAccess_isRejected`	Attempts to access `GET /api/teachers` and `POST /api/teachers` without login. Confirms that Spring Security halts the request and returns **401 Unauthorized**.

Code Walkthrough: Full Lifecycle Integration Test

```
// TeacherIntegrationTest.java
@Test
void fullTeacherLifecycle_createReadUpdateDelete() throws Exception {
    // 1. Create Teacher as Admin
    MvcResult result = mvc.perform(post("/api/teachers")
                    .with(httpBasic("admin", "admin123"))
                    .contentType(MediaType.APPLICATION_JSON)
                    .content(TEACHER_PAYLOAD))
            .andExpect(status().isCreated())
            .andExpect(header().string("Location", containsString("/api/teachers/")))
            .andExpect(jsonPath("$.password").doesNotExist()) // Never leak hash
            .andReturn();

    String location = result.getResponse().getHeader("Location");

    // 2. Read created Teacher
    mvc.perform(get(location).with(httpBasic("admin", "admin123")))
            .andExpect(status().isOk())
            .andExpect(jsonPath("$.email").value("hasan@uni.edu"));

    // 3. Delete Teacher
    mvc.perform(delete(location).with(httpBasic("admin", "admin123")))
            .andExpect(status().isNoContent());

    // 4. Verify Not Found
    mvc.perform(get(location).with(httpBasic("admin", "admin123")))
```

```
            .andExpect(status().isNotFound());
}
```

> 💡 **Viva / Teacher Defense Tip #2: How are test databases kept clean without leftover data?**
>
> **Teacher Question:** *"If test 1 inserts a teacher, won't that interfere with test 2? How do you prevent data collision?"*
>
> **Your Answer:** *"We annotate the test class with Spring's `@Transactional`. In a test context, Spring automatically rolls back the transaction at the end of every individual test method. The data is written to the H2 in-memory database during the test and immediately discarded, ensuring 100% test isolation."*

4. Continuous Integration Pipeline (GitHub Actions)

The CI pipeline ensures that any commit pushed to the repository is immediately checked out, built, and tested inside a clean container environment. If any unit or integration test fails, the build fails and alerts the developer.

Workflow Configuration: `.github/workflows/ci.yml`

```yaml
name: Java CI with Maven

on:
  push:
    branches: [ "main" ]
  pull_request:
    branches: [ "main" ]

jobs:
  build:
    runs-on: ubuntu-latest

    steps:
    - name: Check out repository
      uses: actions/checkout@v5          # Updated to v5 (Node 24 runtime)

    - name: Set up JDK 21
      uses: actions/setup-java@v5        # Updated to v5 (Node 24 runtime)
      with:
        java-version: '21'
        distribution: 'temurin'
        cache: maven                     # Caches ~/.m2 to speed up builds

    - name: Make Maven wrapper executable
      run: chmod +x mvnw

    - name: Run tests with Maven
      run: ./mvnw -B test                # Batch mode: headless, deterministic
```

> ⚠️ **Engineering Note: Resolving Node.js 20 Deprecation Warnings**
>
> Earlier GitHub Actions used `v4` of `checkout` and `setup-java`, which emitted deprecation warnings because GitHub runners are phasing out the Node.js 20 runtime. We upgraded both actions to `v5`, which runs natively on **Node.js 24**. As a result, our CI builds run with **zero warnings**.

5. Experimental Results & Verification

When running `./mvnw test` locally or via GitHub Actions, all **25 test cases** across the test suite execute and pass with zero failures:

```
[INFO] -------------------------------------------------------
[INFO]  T E S T S
[INFO] -------------------------------------------------------
[INFO] Running com.university.ums.UniversityApiTests
[INFO] Tests run: 16, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 5.434 s
[INFO] Running com.university.ums.service.StudentServiceTest
[INFO] Tests run: 3, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.167 s
[INFO] Running com.university.ums.service.AdminServiceTest
[INFO] Tests run: 3, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.049 s
[INFO] Running com.university.ums.TeacherIntegrationTest
[INFO] Tests run: 3, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.783 s
[INFO]
[INFO] Results:
[INFO] Tests run: 25, Failures: 0, Errors: 0, Skipped: 0
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
```

Local vs. Remote CI Benchmark Comparison

Metric	Local Machine (macOS ARM64)	GitHub Actions Runner (Ubuntu 24.04 LTS)
Total Test Count	25 tests	25 tests
Unit Test Duration (6 tests)	**0.216 s** (~36 ms / test)	**0.280 s** (~46 ms / test)
Integration Test Duration (3 tests)	**0.783 s**	**0.755 s**
Full Suite Test Time	6.43 s	6.95 s
End-to-End Build / Job Time	7.55 s	24 s – 32 s (includes checkout, JDK setup & caching)
Exit Code & Status	`BUILD SUCCESS (0)`	`success (0)`

6. Key Discussion Points for Lab Evaluation

1. **Test Pyramid Economics:** Unit tests execute in **0.22 seconds**, whereas integration tests take **6+ seconds** due to container boot time. This proves why comprehensive branch coverage belongs in unit tests, while integration tests verify contracts and security.
2. **Security Verification:** Testing both authenticated and unauthenticated scenarios guarantees that Spring Security filters cannot be bypassed. The integration tests confirm that passwords are never returned in JSON responses and that passwords are stored as BCrypt hashes starting with `$2a$`.
3. **Deterministic Automation:** By committing `mvnw` (Maven Wrapper) and specifying exact JDK 21 distribution in GitHub Actions, the build is completely reproducible regardless of the operating system of individual developers.

7. Quick Presentation Cheat Sheet (1-Minute Teacher Demo)

> **If you have 1 minute to explain this to your teacher, say this:**
>
> *"Sir/Ma'am, for this project I implemented a complete automated testing and CI/CD strategy. First, I created **6 isolated unit tests** using JUnit 5 and Mockito to test our service layer rules—like preventing anyone from deleting the last admin, and blocking duplicate student IDs—running in just 0.2 seconds. Second, I created **3 full-slice integration tests** with Spring MockMvc and an in-memory H2 database to verify the complete CRUD lifecycle, password hashing confidentiality, and 401/409 error handling across the security filters. Finally, I set up a **GitHub Actions CI pipeline** with JDK 21 that automatically triggers on every push and pull request, caches dependencies, and guarantees all 25 tests pass green before code is accepted."*